Quiz: adding an item to a stack, a queue and a linear linked list

20 minutes · 20 marks · start of the next lesson, before pop, dequeue and delete

Name		Class		Score	/ 20

Rules. Arrays are numbered from 1. The value 0 means empty for TOP and REAR, and null for NEXT. An empty queue has FRONT = 1 and REAR = 0. Work alone: no notes, no visualiser, no chatbot. Explain answers from the arrays and arrows.

Section A · Multiple choice (6 marks)

Circle one letter for each question. 1 mark each.

A1. A stack is stored in S[1..6]. Now S[1] = 5, S[2] = 18, S[3] = 22 and TOP = 3. After **push 9**, which of the following is correct?

A. TOP = 4 and S[4] = 9
B. TOP = 3 and S[3] = 9
C. TOP = 4 and S[1] = 9
D. TOP = 2 and S[4] = 9

A2. One item is added to each structure: a push onto a stack, an enqueue onto a queue, and an insert in the **middle** of a linear linked list. Which of these pointers change?
(1) TOP of the stack
(2) FRONT of the queue
(3) HEAD of the linked list

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

A3. A queue is stored in Q[1..6]. Now Q[1] = 33, Q[2] = 47, FRONT = 1 and REAR = 2. After **enqueue 60**, which of the following is correct?

A. FRONT = 2, REAR = 3, Q[3] = 60
B. FRONT = 1, REAR = 3, Q[3] = 60
C. FRONT = 1, REAR = 2, Q[2] = 60
D. FRONT = 1, REAR = 3, Q[1] = 60

A4. A linear linked list is stored in the arrays below. HEAD = 1 and NEXT = 0 means null. Starting from HEAD, in what order are the values in the list?

Index	DATA	NEXT
1	40	3
2	12	0
3	27	2
4	—	0

A. 40, 12, 27
B. 27, 12, 40
C. 12, 27, 40
D. 40, 27, 12

A5. In a linear linked list, NEXT[1] = 2. The new value is already stored in the free node, DATA[4]. Which two statements, **in this order**, insert node 4 between node 1 and node 2?

- **A.** NEXT[1] ← 4, then NEXT[4] ← NEXT[1]
- **B.** NEXT[4] ← 4, then NEXT[1] ← NEXT[4]
- **C.** NEXT[4] ← NEXT[1], then NEXT[1] ← 4
- **D.** NEXT[1] ← NEXT[4], then NEXT[4] ← 1

A6. Which of the following statements about **adding** an item is / are correct?
(1) In a stack, TOP increases by 1 before the new value is written into S[TOP].
(2) In a linear linked list, the new value is written into a free node.
(3) In a linear linked list stored in arrays, inserting a value in the middle moves every later value one cell to the right.

- **A.** (1) only
- **B.** (1) and (2) only
- **C.** (2) and (3) only
- **D.** (1), (2) and (3)

Section B · Programs and traces (14 marks)

Blue cells are given. Write every yellow cell. Leave a cell blank if it stays empty.

B1. Stack · dry run a program (4 marks)

A stack is stored in S[1..6]. Before the program runs, the stack holds four items and TOP = 4, as shown on the right.

Program

```
1   for I ← 1 to 3
2      X ← I * 10
3      if TOP = 6 then
4         output “stack overflow”
5      else
6         TOP ← TOP + 1
7         S[TOP] ← X
8      endif
9   next I
10  output TOP
```

Stack S (top of the drawing = S[6])

Index	S
6	
5	
4	5
3	17
2	3
1	8

TOP =

(a) Complete the stack on the right after the program ends, and write the final value of TOP. [2]

(b) Write everything the program outputs, in order. [1]	

(c) One value of X is never stored. Which value, and why? [1]

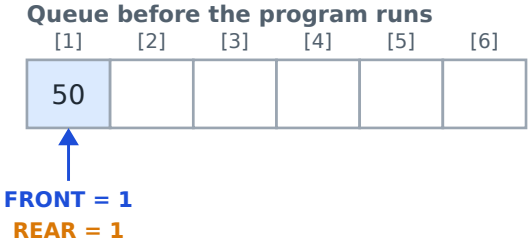
B2. Queue · trace table (3 marks)

A queue is stored in Q[1..6]. Before the program runs, Q[1] = 50, FRONT = 1 and REAR = 1. The array IN holds IN[1] = 62, IN[2] = 73 and IN[3] = 81.

Program

```
1  for I ← 1 to 3
2     if REAR = 6 then
3        output “queue overflow”
4     else
5        REAR ← REAR + 1
6        Q[REAR] ← IN[I]
7     endif
8  next I
9  output Q[FRONT], Q[REAR]
```

Queue before the program runs



(a) Complete the trace table. Write the values **at the end of** each pass of the loop. [2]

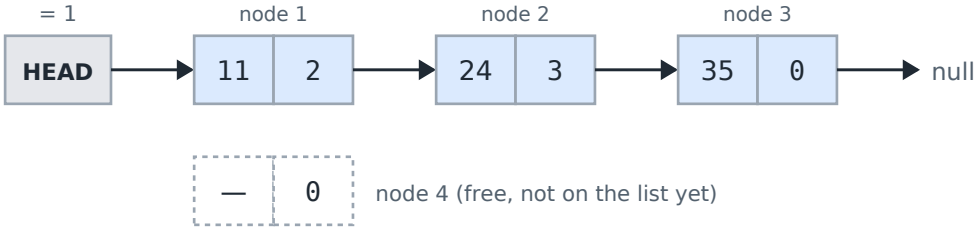
I	IN[I]	FRONT	REAR	Cell written
1	62			
2	73			
3	81			

(b) Write the output of line 9. [1]

B3. Linear linked list · find and correct the logic error (7 marks)

HEAD = 1. The list is 11 → 24 → 35 → null. Node 4 is free. The aim is to insert **30 between 24 and 35**.

The list before the insert



Each node: DATA | NEXT. NEXT = 0 means null.

Arrays before the insert

Index	DATA	NEXT
1	11	2
2	24	3
3	35	0
4	—	0

A student wrote this program to do the insert

```
1  DATA[4] ← 30
2  NEXT[2] ← 4
3  NEXT[4] ← NEXT[2]
```

(a) Dry run the student's program. After it ends, what are NEXT[2] and NEXT[4]? [2]

NEXT[2] =		NEXT[4] =	

(b) Which value can no longer be reached from HEAD, and why? [1]

(c) Rewrite the program so the insert works. [2]

```
1
2
3
```

(d) Complete the arrays after **your** program in (c) runs. [2]

Index	DATA	NEXT
1		
2		
3		
4		

End of quiz. Hand in this sheet.